NITSHACKS 9.0

IDEA SUBMISSION

NITSHACKS 9.0

North-east India’s Most Premium Hackathon

PROBLEM STATEMENT TITLE

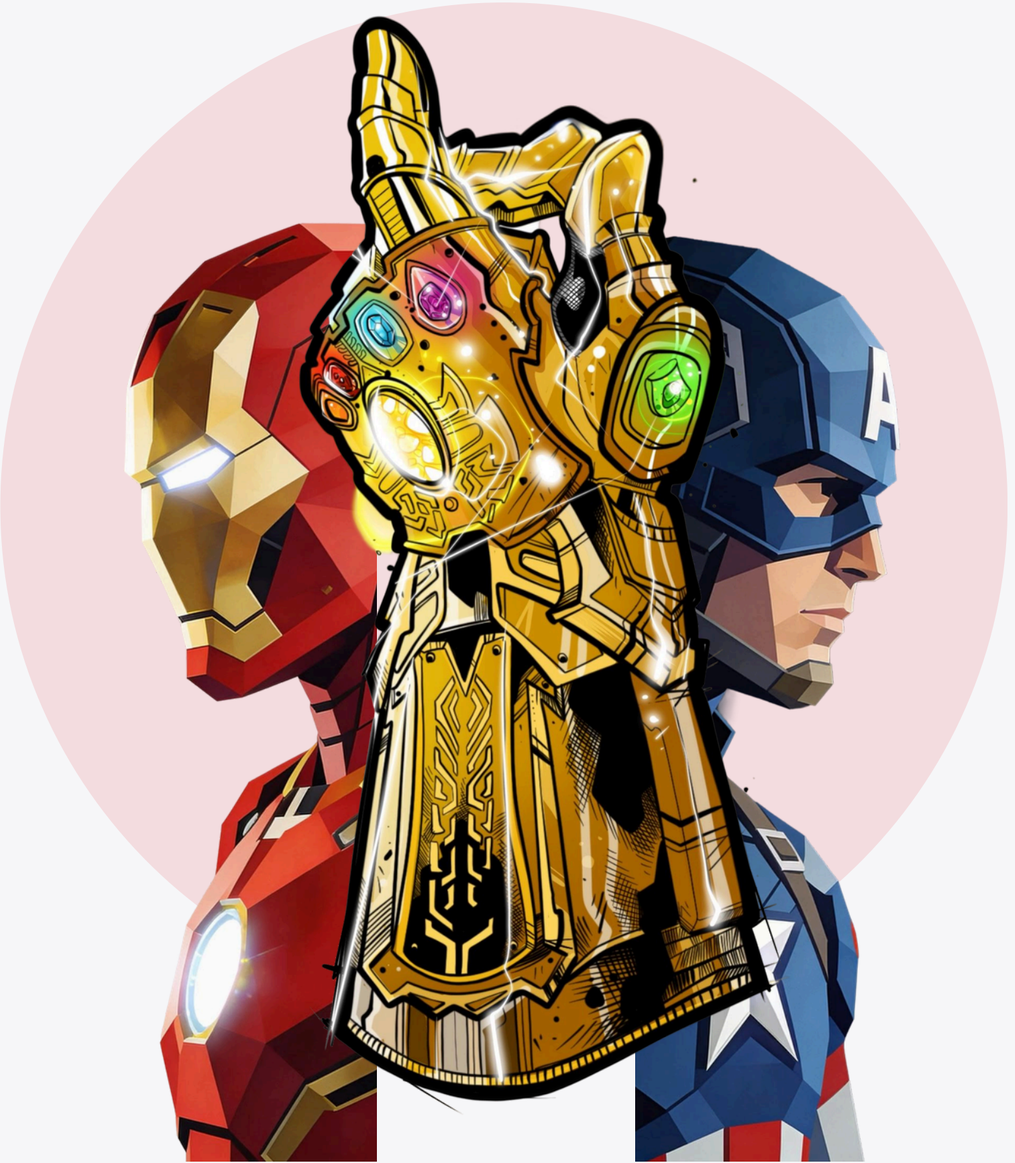
J.A.R.V.I.S. - Justified Analysis, Reasoning & Visualization of Incident Systems

THEME

Artificial Intelligence, Intelligent Systems & Distributed Systems

TEAM NAME (REGISTERED ON PORTAL)

AquaNox



01 / PROPOSED SOLUTION

J.A.R.V.I.S -
Autonomous Incident Investigation and Self-Healing SRE Platform

Proposed Solution

In distributed systems, alerts often arrive from multiple connected services. Engineers must manually correlate events, identify root causes, understand impact, and recover the system - leading to longer downtime.

Our solution:



What Makes J.A.R.V.I.S AI Innovative & Unique?

1. **AI-Powered Root Cause Analysis**: Intelligently analyzes alerts, metrics, and events to identify the most probable cause of system failures.
2. **Intelligent Event Correlation**: Connects related alerts across multiple microservices into a single incident, reducing duplicate alerts and alert fatigue.
3. **Automated Impact Analysis & Remediation**: Identifies affected services, visualizes dependencies, and recommends or performs safe recovery actions.
4. **Self-Healing with Recovery Verification**: Verifies system health after remediation to confirm successful recovery and reduce Mean Time to Recovery (MTTR).

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02 / BUILD PLAN

TECHNICAL APPROACH

Technologies to be used

NEXT.JS

Ollama

FastAP

LangGraph

PostgreSQL

docker

Methodology & implementation

1. Microservices generate metrics, logs and events; log files can also be uploaded.
2. All events are normalized into one unified schema.
3. Monitoring rules detect abnormal behavior and raise alerts.
4. Correlation groups related events by time window, dependencies and shared services.
5. Root cause analysis returns evidence and a confidence score; blast radius is estimated.
6. The Governance Agent checks the fix, then a safe action runs.
7. Recovery is verified with health checks and metrics.
8. Dashboard shows incident status, timeline, dependency graph and post-mortem.

Microservices + uploaded logs
Gateway · User · Order · Payment · Notification

↓

Prometheus + AlertManager
Metrics collection and alerts

↓

Event schema + correlation
Time window · dependencies · shared service

↓

LangGraph agent
Root cause · impact · governance

↓

Remediation + verification
Restart · rollback · scale · health checks

03 / REALITY CHECK

FEASIBILITY AND VIABILITY

Feasibility

- Uses open-source tools and a local AI model, reducing initial infrastructure costs.
- Mock microservices allow controlled testing without a production environment.
- SQLite supports local development; PostgreSQL supports a more persistent deployment.
- Modular services make future integration with real infrastructure possible.

Challenges & Risks

Too many or duplicate alerts
Incorrect AI root-cause predictions
Unsafe automated actions
False recovery signals
Local model performance limitations
Data loss or service failure

Mitigation

Correlation rules, grouping, and time-window analysis
Use metric evidence, dependency data, and confidence scores
Policy checks, approval for risky actions, and action limits
Verify health checks, error rates, latency, and service availability
Keep workflows lightweight and use deterministic checks alongside AI
Database backups, logs, and controlled rollback procedures

04 / WHY IT MATTERS

IMPACT AND BENEFITS

Target Audience

- DevOps and Site Reliability Engineering (SRE) teams.
- Cloud and infrastructure engineers.
- Software development and operations teams.
- Startups and organizations operating microservices.
- IT support and incident response teams.

Expected Impact

- Faster incident investigation and recovery.
- Reduced alert fatigue through event correlation.
- Better visibility into root causes and affected services.
- Consistent incident timelines and postmortem reports.
- Safer, more structured remediation workflows.

Benefits



05 / EVIDENCE

RESEARCH AND REFERENCES

What backs up your idea?

Research

Anomaly Detection and Root-Cause Identification in Microservices: A Survey (2026)
A research survey covering AI-based anomaly detection, fault localization, and root-cause analysis in microservices.

Click here

GALA: Graph-Augmented LLM Agents for Root Cause Analysis and Incident Response in Microservices (2026)
Research on combining LLM agents with service dependency graphs for incident investigation and response.

Click here

References

Debugging Incidents in Google's Distributed Systems (2020)
Explains how engineers investigate and debug complex production incidents in distributed systems.

Click here

Google SRE Book - Monitoring Distributed Systems
Covers monitoring, alert quality, symptoms versus root causes, and incident detection.

Click here